Traveling Shots in Language: Towards an Analysis of Dynamic Viewpoints in ASL*

Philippe Schlenker[a], Jason Lamberton[b], Jonathan Lamberton[c]

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Abstract. In sign language, classifier predicates have a highly iconic semantics, which has motivated the development of a formal framework ('Iconological Semantics') with a pictorial component evaluated relative to viewpoint variables. But in this framework, viewpoints are static. We argue that this analysis fails to capture a well-documented phenomenon in several sign languages: sometimes classifier movement (e.g. of a tree-denoting classifier) is used to represent the apparent motion of a static object relative to a moving viewpoint, a narrative technique corresponding to traveling cameras in film. To address this problem, we propose that viewpoint variables may denote dynamic (traveling) viewpoints. We consider two analyses: one in which all viewpoints can be dynamic; the other in which only viewpoints introduced by Role Shift (analyzed as overt context shift) can be. For lack of a consensus on the definition of Role Shift, both options are currently open.

Keywords: sign language, ASL, semantics, iconological semantics, viewpoints, dynamic viewpoints, iconicity, pictorial semantics, visual narratives

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* **Authors' contributions:** Philippe Schlenker initiated this research, and developed the analysis in discussion with Jonathan Lamberton. All examples were constructed by and with Jonathan Lamberton, who also played a key role in formulating the correct generalizations and in exploring possible counterexamples. In particular, the possibility that incomplete Role Shift was involved in many examples was highlighted by Jonathan Lamberton. After the article was mostly written, Jonathan Lamberton met with his brother Jason Lamberton to record the latter's judgments (in particular by comparison with his own).

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[a] Institut Jean-Nicod (ENS - EHESS - CNRS), Département d'Etudes Cognitives, Ecole Normale Supérieure, Paris, France; PSL Research University.
[b] Motion Light Lab, VL2, Gallaudet University, USA.
[c] New York, NY, USA.

1 Introduction

Sign languages have the same general grammatical and semantic properties as spoken languages (e.g. Sandler and Lillo-Martin 2006), but they notoriously make greater use of iconicity (e.g. Emmorey 2023). One of the most studied cases involves classifier predicates ('classifiers' for short, also called 'classifier constructions'). They have a lexically specified handshape, standing for a class of referents, but their position in signing space, their orientation and their movement (if there is one) are unconstrained, and interpreted iconically (according to Zwitserlood 2012, classifier predicates "are reported to occur in almost all sign languages researched to date"; see also an early characterization in Supalla 1982). Just as would be the case for a picture or a video, classifier predicates must be interpreted relative to a viewpoint, corresponding intuitively to the position of the camera. In standard cases, the viewpoint is static, and movement of the classifier indicates that its denotation (for instance a car) is in motion. But several authors have noted a striking phenomenon in which classifiers that denote static objects, such as a tree, can still move in signing space to represent relative motion as perceived by a moving viewpoint (e.g. Lucas and Valli 1990 and Liddell 2003 for ASL [American Sign Language]; Müller 2018 for DGS [German Sign Language]; Cuxac 2007 for LSF [French Sign Language]). In effect, this means that the cinematic technique of traveling shots is instantiated in language (Liddell 2003, Müller 2018; see also Parrill et al. 2016).[1] But how should this phenomenon be formally captured?

We start from a recent analysis of iconic viewpoints, developed to offer a synthesis (called 'Iconological Semantics') of compositional and iconic semantics in sign language and beyond (Schlenker and Lamberton 2024). We argue that a correction is needed. In its original version, this framework was based on viewpoint variables denoting static viewpoints. We propose that viewpoints may be dynamic and thus give rise to a relative motion interpretation. We consider two theories. According to one, all viewpoints can in principle be dynamic. According to the other, only viewpoints introduced by Role Shift (an operation sometimes analyzed as overt context shift) can be. For lack of a consensus on the definition of Role Shift, both options are currently open: examples without the standard hallmarks of Role Shift could be reanalyzed as involving incomplete versions of it, hence an open question for future research.

Data were obtained from two Deaf native signers of ASL, who are also co-authors of this piece. Details about elicitation methods and transcription conventions (which are standard) appear in Appendix A. We write acceptability on a 7-point scale as a superscript at the beginning of sentences, with 7 = best and 1 = worst. Anonymized videos are systematically linked to the text to help specialists follow examples (the videos can also be found in the Supplementary Materials).

2 Viewpoint Shift and Role Shift

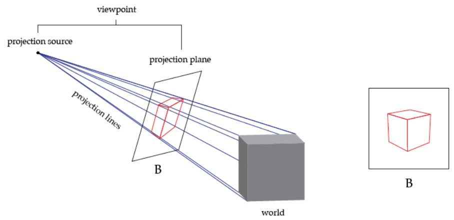
2.1 Viewpoint Shift

To capture the interaction between classifiers and compositional semantics, Schlenker and Lamberton 2024 propose a framework that combines standard logical semantics with formal pictorial semantics in the tradition of Greenberg 2013, 2021 and Abusch 2013, 2020. Schlenker and Lamberton's proposal, Iconological Semantics, builds on the observation that pictorial semantics provides a definition of truth for pictures relative to a time, a world, an assignment function, and a viewpoint. In a Greenbergian framework, a picture is true at a time t in a world w if at t the world projects to the

[1] Since our data are based on elicitation, not naturalistic use, we not seek to determine the type of register in which this phenomenon can be found.

picture in view of relevant projection method. For example, in (1), a cube projects onto the picture (the projection plane) according to rules of perspective projection. But this requires the specification of a viewpoint, in this case of a projection source and of the projection plane.

(1) **An example of a projection method: perspective projection (Greenberg 2021)**



To this framework with viewpoints, times and worlds, Abusch 2013, 2020 proposed to add a dependency on assignment functions. Her motivation was that in silent pictorial narratives, there are ambiguities of coreference that are elegantly handled by positing covert variables, and hence assignment functions (see also Schlenker and Lamberton 2024 for discussion).

Evaluation relative to a time, world and assignment function is what standard compositional semantics offers, and thus only viewpoint dependency requires a special treatment to allow for a unification of logical and pictorial semantics. In Iconological Semantics, classifiers and other iconic expressions come with a viewpoint variable π_i (different variables for different values of *i*), which simultaneously tells the semantic system that the expression has a pictorial component, and specifies the viewpoint relative to which it must be interpreted. To illustrate, we can start from the lexical entry for a standard nominal (not a classifier), *PLANE,* in (2a). The minimally different airplane classifier in (2b), *PLANE-cl*, has this lexical component as well, but in addition it has a projective component (boldfaced). Here $proj(x, s(\pi_i), t, w) = plane\text{-}cl$ means that at t in w, from viewpoint $s(\pi_i)$, x projects to the classifier *plane-cl*.[2]

(2) a. $[[PLANE]]^{c, s, t, w} = \lambda x_e . plane'_{t, w}(x)$
b. $[[PLANE\text{-}cl^{\pi\_i}]]^{c, s, t, w} = \lambda x_e . plane'_{t, w}(x) = 1$ and $\mathbf{proj(x, s(\pi_i), t, w) = plane\text{-}cl}$

In essence, an object x is in the extension of $PLANE\text{-}cl^{\pi}$ relative to assignment function s at time t in world w just in case, at t and in w, x is a plane, and in addition (= boldfaced part), x projects to the position and orientation specified by the classifier relative to viewpoint specified by the viewpoint variable π, namely $s(\pi)$.

Viewpoint variables allow for considerable flexibility in the interaction between the logical and the pictorial components. In some cases, a single viewpoint variable appears and is free, which means that its value (the 'salient viewpoint') is provided by the context. In other cases, viewpoint variables can be bound by an existential quantifier, which can take scope under further operators. In still other cases, as illustrated in (3a), with the highly simplified Logical Form in (3b), two viewpoint variables can coexist in the same sentence.[3]

(3) *Context*: The signer teaches a philosophy class and a linguistics class in different buildings.
a. [7] HAVE ONE STUDENT [PHILOSOPHY CLASS]$_a$ BREAK ALWAYS a-PERSON-WALK-slow-cl-1, [LINGUISTICS CLASS]$_b$ BREAK ALWAYS b-PERSON-WALK-slow-cl-1

[2] The fact that the classifier version resembles the nominal version should not give the wrong idea: in Schlenker and Lamberton's (2024) system, there is no derivational relationship between the two, and no claim either that under certain conditions nominals can be 'turned into' classifiers.
[3] As noted in fn. 22, the assumption that viewpoints are functional makes it possible to re-analyze some of these examples without viewpoint variables. See Schlenker and Lamberton 2024, Appendix IC and ID.

'There is one student who during philosophy class breaks always slowly approaches the teacher from the right and during linguistics class breaks always slowly approaches the teacher from the left .'
(ASL, 35, 2228a, 3 judgments; https://youtu.be/IXP4iPPsoZU, 1st sentence; Schlenker and Lamberton 2024)

b. there-is student λi [always$_T$ t_i person-walk-cl$^{\pi_i}$ and always$_{T'}$ t_i person-walk'-cl$^{\pi_k}$]

The first classifier, analyzed as *person-walk-cl*$^{\pi\_i}$, represents the relevant student approaching the teacher from the left in the philosophy class, hence a viewpoint variable π_i; the second classifier, analyzed as *person-walk'-cl*$^{\pi\_k}$, represents the same student approaching the teacher from the right in a different classroom, hence a different viewpoint variable π_k.

Further adjustments are needed to incorporate pictorial semantics into a linguistic semantics. The Greenberg/Abusch pictorial semantics is defined in terms of 2D geometric projections, but as discussed by Schlenker and Lamberton, it must be adapted to the case of 3D representations used in sign language. As a result, the notion of a viewpoint is modified as well. In a nutshell, for Schlenker and Lamberton, the viewpoint is the real-world observation point from which a scene is observed and then recreated by the signer in a scaled version by way of puppet-like elements (classifiers), as illustrated in (4). Schlenker and Lamberton take the viewpoint to include a Cartesian coordinate system, relative to which the position and orientation of real-world objects can be specified. In addition, Schlenker and Lamberton refine the analysis so as to account for moving classifiers, a point to which we return below.

(4) **Obama sitting across from the Dalai Lama: from the scene to its classifier representation**[4]

a. Original scene, with the viewpoint (☺) behind the Dalai Lama

b. The signer recreates a scaled version of the 3D scenewith two classifiers representing Obama and the Dalai Lama

In Iconological Semantics, viewpoints were taken to be static; in the case of pictures, these would correspond to a static 'camera position' relative to which the depictive component of classifiers is assessed. But we will discuss data showing that sometimes viewpoints are in movement, which necessitates an extension of this framework: viewpoint variables will have to denote dynamic viewpoints, or in other words functions from times to static viewpoints.

2.2 Role Shift

There is another sign language construction that intuitively involves viewpoints, namely Role Shift. In Role Shift, the signer signals that he adopts another character's perspective, and this has various grammatical and semantic consequences. Action Role Shift (also called Constructed Action) pertains to action reports, Attitude Role Shift (also called Constructed Dialogue) to attitude reports (see Metzger 1995, Schlenker 2017a,b).[5] But a key issue pertains to the definition of Role Shift. In a broad sense,

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[4] From Schlenker and Lamberton 2024. Picture credit (Obama and the Dalai Lama): Pete Souza, Public domain, via Wikimedia Commons (we use a mirror-image version).
[5] See also Kegl 1986, 1987 for relevant early formal work (Role Shift seems to correspond to what she calls "role prominence").

whenever the signer's body (and/or facial expressions) represents some other character, Role Shift is involved; in this sense, Role Shift is closely connected to "character viewpoint" gestures, which help "reenact the character" (McNeill 1992; see Cormier et al. 2012, Quinto-Pozos and Parrill 2015, Parrill et al. 2016, Waller 2021). In a stricter sense, usually adopted in the formal literature, Role Shift is characterized by designated properties, notably: (i) 'temporary interruption of eye contact with the actual interlocutor'; (ii) a 'slight shift of the upper body in the direction of the locus associated' with the relevant character; (iii) a 'change in head position'; (iv) a 'facial expression associated' with the relevant character (Quer 2013).

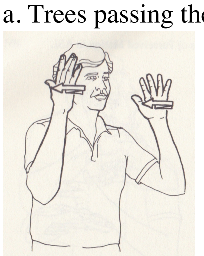
ASL Role Shift is usually thought to have additional grammatical properties: (v) for verbs that display subject agreement, if agreement is with the character whose perspective is adopted, agreement targets the signer's role-shifted position (e.g. Frederiksen and Mayberry 2018 Figure 3); and (vi) indexicals such as first person elements cannot refer to the signer under Role Shift (e.g. Schlenker 2017a,b; we partly challenge this point in Appendix B). In addition, in ASL (as well as in LSF), (vii) Role Shift has been shown to come with a strong iconic component (Schlenker 2017b; see also Davidson 2015), and (viii) Role Shift intuitively comes with the iconic viewpoint of the character whose perspective the signer adopts (e.g. Cormier et al. 2012, Quinto-Pozos and Parrill 2015, Waller 2021).

The examples we constructed came in two versions, one with and one without the hallmarks of Role Shift in the strict sense. Correspondingly, we will develop two analyses, one for dynamic viewpoints associated with standard viewpoint variables, and another for dynamic viewpoints associated with Role Shift. But if the broad notion of Role Shift is adopted, all our examples might involve Role Shift. If so, one may conclude that only our second analysis is needed: all dynamic viewpoints must be introduced by Role shift, while standard viewpoint variables denote static viewpoints.

3 Classifiers representing relative motion in sign languages: earlier work

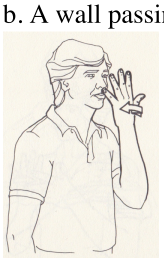
In interviews with three native signers of ASL, Lucas and Valli 1990 found several naturalistic examples involving classifiers representing relative motion in descriptions of eight clips of James Bond movies. Example were of two types. In one, the signer's body represents a moving character[6], and classifiers represent static objects (relative to the earth) that appear to be moving from this character's perspective: trees pass the character on both sides in (5a), and a wall passes in front of the character in (5b). Both examples involve the open hand classifier, used, among others, to represent large flat objects and trees.

(5) a. Trees passing the character on both sides



(Lucas and Valli 1990 Fig. 7.20 p. 162)

b. A wall passing in front of the character



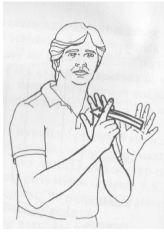
(Lucas and Valli 1990 Fig. 7.21 p. 162)

In another type of examples, a classifier representing a static object moved relative to a classifier representing a moving object: a tree (left hand) passing a person

[6] Lucas and Valli 1990 use the term "signer" rather than "character", but it's unlikely (in view of the James Bond context) that the consultants would have talked about themselves.

(right hand), as in (6a), or a surface (left hand) passing under a vehicle going downhill (right hand), as in (6b).

(6) a. A tree (left hand) moving past a person (right hand) b. A surface (left hand) passing under a vehicle going downhill (right hand)



(Lucas and Valli 1990 Fig. 7.8 p. 158)

(Lucas and Valli 1990 Fig. 7.11 p. 159)

It is clear from Lucas and Valli's description that classifier movement can be modulated in highly iconic ways, as is expected of classifiers in general. Similar example types were mentioned by Liddell 2003 (pp. 296-301). Müller 2018 discusses numerous related examples in German Sign Language (DGS), from the perspective of the application to sign language of concepts from filmmaking, in particular that of camera movement ("Kamerabewegung"), which is immediately relevant here.

Restricting attention to ASL, Lucas and Valli's description does not include full transcriptions, which makes it hard to assess what triggers the use of dynamic viewpoints. In particular, it is difficult to determiner whether these examples involve Role Shift.

4 Relative motion in ASL: elicited examples without Strict Role Shift

To obtain more complete and controlled paradigms, we elicited examples involving a single classifier moving relative to the signer, who represents a character. Since in Schlenker and Lamberton 2024 the argument for Viewpoint Shift does not rely on Role Shift, we focused on examples that did not have the characteristic hallmarks of Role Shift in the strict sense. Specifically, (i) in all examples discussed in this section, there was no body rotation. In addition: (ii) in some (namely (16)) there was no eyegaze shift;[7] (iii) some examples involved subject agreement verbs that would have been expected to originate in the signer's position under Role Shift, but didn't; (iv) some examples involve first person pronouns, which cannot normally refer to the signer under Role Shift (e.g. Schlenker 2017a, but we adduce conflicting data in Appendix B). In constructing paradigms, we paid close attention to the presence vs. absence of body rotation because, in our experience, it is a rather unambiguous marker of Role Shift (which does not entail that its *absence* signals lacks of Role Shift).[8] Still, in each example, the signer's body represented another character, which means that Role Shift in the broad sense was involved.

Our simplest paradigm appears in (7) (here $_{\text{clumsy}}$ transcribes a clumsy facial expression). It features a pole classifier moving towards the signer, who adopts the viewpoint of a character, Ann.

(7) POSS-1 HOUSE HAVE POLE-a. SOMETIME CAR HIT-a. YESTERDAY ANN$_b$ SELF-b DRUNK.
'My house has a pole that cars sometimes hit. Yesterday, Ann was drunk, and

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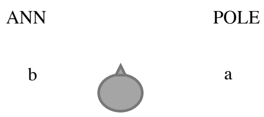
[7] There was eyegaze shift on the first person possess in (7) and (10). There was quick eyegaze shift on *IX-a* in (13).

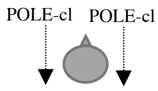
[8] As a reviewer notes, in German Sign Language (DGS), it was argued Herrmann and Steinbach 2012 that a minimal marker of Role Shift is eyegaze shift. As we explained at the outset, the status of incomplete versions of Role Shift (with some but not all characteristic features) is a crucial open question to assess present results.

IX-b
she
a. clumsy____
<RS__>
[7] DRIVE POSS-1 POLE POLE-passing_right-cl_.
nearly hit my pole on her right while driving.
b. clumsy____
<RS__>
[7] DRIVE POSS-1 POLE POLE-passing_left-cl.
nearly[9] hit my pole on her left while driving.
(ASL 37, 3404) https://youtu.be/hCPu9_F5dvY

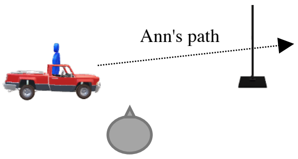
In the first three sentences, the pole nominal is introduced on the signer's right, and Ann on the signer's left, as illustrated in (8a). In the last sentence, the pole classifier moves past the signer's head on the right (in (7a)) or on the left (in (7b)), as illustrated in (8b), with different inferences about Ann's path, as shown in (9).

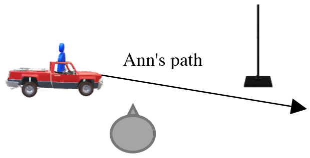
(8) a. Loci for Ann and the pole nominal b. Movement of the classifier representing the pole passing by in (7a) (right) and (7b) (left)





(9) a. Ann's inferred path in (7a) b. Ann's inferred path in (7d-e)





It is clear that the signer adopts Ann's viewpoint, and his facial expressions on *DRIVE* evoke Ann's drunken attitude. Although there is no body shift, it cannot be excluded that *DRIVE* is role-shifted (owing to the facial expressions), hence our use of *RS* in angle brackets in (7). The pole classifier moves towards the signer, indicating that the viewpoint is now moving. Depending on whether the pole classifier moves past the signer's head on the left or on the right, it is understood that the character, Ann, nearly hit the pole on her right (path in (9a)) or on her left (path in (9b)). Importantly, the object of *DRIVE* involves the first person possessive pronoun, which denotes the signer, something that is unexpected if the object is under Role Shift. However, as we discuss in Appendix B, there are exceptions to this generalization in our data, and this point must thus be interpreted with caution.

The same point is made with three different classifiers representing a house or house corner in (10).

(10) YESTERDAY IX-1 SEE POSS-1 HOUSE-a. ANN SELF-b DRUNK. IX-b DRIVE
'Yesterday I saw my house, and nearby Ann, who was drunk. She drove
b-VEHICLE-cl-mid POSS-1 HOUSE$_a$ mid-VEHICLE-cl-a-a$_{front}$
towards my house and took a left, nearly missing

a. [7] WALL-passing_right-cl_.
my house on her right.'

b. [6.7] RECTANGLE-passing_right-cl_.

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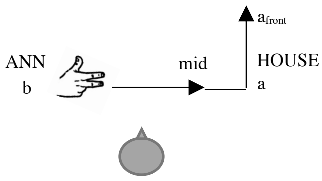
[9] Our second consultant initially understood the sentence as meaning that the car hit the pole.

my house on her right.'

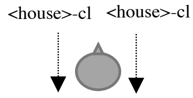
c. [7] CORNER-passing_right-cl_.
the corner of my house on her right.'
d. [6] WALL-passing_left-cl.
my house on her left.'
e. [6.3] RECTANGLE-passing_left-cl.
my house on her left.'
f. [6.3] CORNER-passing_left-cl.
the corner of my house on her left.'
(ASL 37, 3380; 3 judgments) https://youtu.be/opGIM9t7Fi4

The beginning of the discourse sets up a locus for Ann on the left and one for the signer's house on the right, and it features a vehicle classifier (namely ) representing Ann's car going from the Ann locus to the house locus and taking a left, as in (11). The classifier movement is in two parts, as seen in the transcriptions and diagrams: *b-VEHICLE-cl-mid* moves from the *b* locus to a *mid* position appearing before the house locus *a*; *mid-VEHICLE-cl-a-a$_{front}$* moves from the mid locus to the house locus *a*, then takes a left to reach the position *a$_{front}$*.

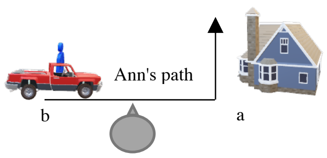
(11) a. Vehicle classifier movement in (10a-f)[10] (left)



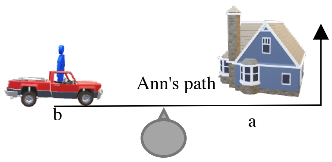
b. Movement of the classifier representing the house passing by in (10a-c) (right) and (10d-e) (<house> stands for different classifiers)



(12) a. Ann's inferred path in (10a-c)



b. Ann's inferred path in (10d-e)



The end of the discourse involves three dynamic classifiers (represented as *<house>-cl* in (11b)) representing a house moving from Ann's viewpoint: an upright open hand, representing a wall, in (10a, d); a two-handed classifier representing a rectangular object in (10b, e); and a two-handed classifier representing a corner in (10c, f). In (10a-c), the classifier moves past the signer's head on his right, whereas in (10d-f) it moves past the signer's head on his left. This triggers different inferences about Ann's path, as represented in (12). This is so despite the fact that the vehicle classifier movement in the preceding sentence is essentially constant across conditions: it is the moving house classifier that is responsible for the inference. Although the signer's head serves to represent Ann's head, characteristic properties of Role Shift in the strict sense are missing; in particular, there is no body shift, and there is a first person possessive pronoun referring to the signer.

The paradigm in (13) makes the same point without a first person possessive pronoun, but with a subject agreement verb, *RUN*, as well as a person classifier, *PERSON,* which both start out from a third person position (if they appeared under Role Shift, they would be expected to start from the signer's position). The first sentence establishes that a person, Ann, was next to a tree. The second sentence describes the relative motion of the tree from a viewpoint associated with Ann.

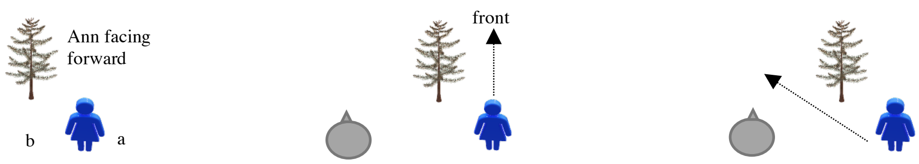
[10] The vehicle classifier representation is extracted from a larger figure in Valli and Lucas 2000.

(13) YESTERDAY IX-1 SEE TREE TREE-cl$_b$. ANN PERSON-cl$_a$, IX-a
'Yesterday I saw a tree. Ann was standing further away. She

a. ^{7} JOG TREE-passing_left-cl_  .
jogged, passing the tree on her left.'
b. $^{6.3}$ a-RUN-front TREE-passing_left-cl.
ran toward the tree, passing it on her left.'
c. ^{5} a-PERSON-front-cl TREE-passing_left-cl.
hurried toward the tree, passing it on her left.'
d. $^{6.3}$ JOG TREE-passing_right-cl.[11]
jogged, passing the tree on her right.'
e. $^{5.3}$ a-RUN-front TREE-passing_right-cl. (5, 7, 4)
ran toward the tree, passing it on her right.'
f. ^{5} a-PERSON-front-cl TREE-passing_right-cl.
hurried toward the tree, passing it on her right.'
(ASL 37, 3126) https://youtu.be/_DixAM4LQVs

(14) a. Loci for Ann and the tree classifier b. Movement of the classifier representing the tree tree passing by in (13a-c) (left) and (13d-f) (right)

(15) a. Position of Ann relative to the tree b. Ann's path displayed by the verbs in (13b,c,e,f), and inferred in (13a) c. Ann's path as inferred in (13d)



The signer uses a tree classifier and a person classifier in the configuration displayed in (14a); this establishes that, relative to the initial viewpoint, the tree was to the left of Ann facing forward, a situation schematically shown in (15a). The next clause involves a verb of running, taking three forms: a plain verb *JOG*; a spatial verb *RUN*, whose initial point agrees with the subject (the third person locus *a*) and whose endpoint agrees with the goal of the movement; and a classifier representing a person moving, also starting from a third person locus *a*. The verb of running is followed by a tree classifier moving towards the signer, on the signer's left in (13a-c), and on the signer's right in (13d-f), as shown in (14b). This yields different inferences about the situation, as illustrated in (15). The spatial verb and the person classifier in (13b, c, e, f) evoke the path displayed in (15b), which is compatible with the tree passing the character on the left, as in (13b, d), but not on the right, as in (13e, f). This explains the deviance of the latter two examples. *JOG* does not display a path and is thus compatible with with trees passing on the left or on the right, but this corresponds to different inferred paths, as illustrated in (15b, c).

In view of the lexical form of *JOG*, which has the signer's arms and head performing the movements of a runner, Role Shift in a broad sense (without rotation) is involved in (13a, d). But this possibility is unlikely in (13b): if Role Shift were involved, we would expect *RUN* and the person classifier to originate from the signer's position, which isn't at all the case here. In this case, Viewpoint Shift without Role Shift seems to be applied. Still, in(13c, e, f), structural similar examples are a bit degraded,

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[11] Our second consultant felt that (13d) is less acceptable than (13d, e). See the Supplementary Materials.

which could suggest that Role Shift is involved.

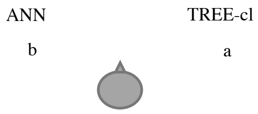
The paradigm in (16) makes related points, but now with Ann hitting the tree. In the first sentence, the same tree classifier as before (glossed as *TREE-cl*) appears on the signer's right, while in the second sentence Ann appears on the signer's left, as shown in (17a). In the last sentence of each example, a different tree classifier, glossed as *TREE-1-cl* (with the 1-handshape, shown in (16a)) moves towards the signer to represent the tree approaching Ann (from her viewpoint) and then hitting her.

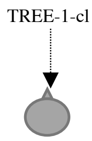
(16) YESTERDAY IX-1 SEE TREE TREE-cl$_a$_SELF PLANT TREE GROW. ANN PERSON-cl$_b$ SELF-b
'Yesterday I saw a tree I had planted myself. Ann
DRUNK. IX-b[12]
was drunk. She

a. $^{6.3}$ JOG TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$_ .
jogged towards the tree and hit her head against it.'
b. $^{5.7}$ b-RUN-a TREE TREE-1-cl-$_{\text{moves and hits the signer's forehead}}$.
ran towards the tree and hit her head against it.'
c. $^{6.3}$ 1-RUN-front$_{\text{signer moves forward}}$ TREE TREE-1-cl-$_{\text{moves and hits the signer's forehead}}$.
ran towards the tree and hit her head against it.'
d. $^{5.3}$ 1-RUN-front$_{\text{signer doesn't move forward}}$ TREE TREE-1-cl-$_{\text{moves and hits the signer's forehead}}$. (7, 4, 5)
ran towards the tree and hit her head against it.'
e. $^{5.7}$ b-PERSON-cl-a TREE TREE-1cl$_{\text{moves and hits the signer's forehead}}$.
hurried towards the tree and hit her head against it.'
f. ^{6} 1-PERSON-cl-front$_{\text{signer moves forward}}$ TREE TREE-1-cl$_{\text{moves and hits the signer's forehead}}$.
hurried towards the tree and hit her head against it.'
g. $^{5.3}$ 1-PERSON-cl-front$_{\text{signer doesn't forward}}$ TREE TREE-1-cl$_{\text{moves and hits the signer's forehead}}$. (7, 4, 5)
hurried towards the tree and hit her head against it.'
(ASL 37, 3142) https://youtu.be/JH3pi-ITXi8

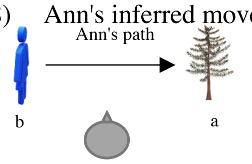
The classifier movement is represented in (17b), and Ann's inferred movement is represented in (18).

(17) a. Loci for Ann and the tree classifier b. Movement of the tree classifier towards the signer





(18) Ann's inferred movement



Different verbs of running appear in different examples: in (16a), *JOG*; in (16b), an agreeing form of *RUN* starting from the Ann locus and ending at the tree locus; in (16c, d) a neutral (non-agreeing) form of *RUN*, moving away from the signer (the signer's body accompanies the movement in c. but not in d). In (16e), a moving person classifier goes from the Ann locus to the tree locus, whereas in (16f, g) the movement is in front of the signer (again with the signer's body accompanying the movement in f but not g). As before, (16b) might make it unlikely that Role Shift is applied, as the agreement verb does not start from the signer's position. But the example

[12] Here our second consultant feels that all examples are "a bit degraded" because the action of hitting the tree should take place at the tree locus not at neutral locus; see the Supplementary Materials. We leave an exploration of this issue for future research.

is a bit degraded, which makes the argument moot.

In sum, we have seen multiple sentences in which classifier movement evokes relative motion. Our examples did not involve the hallmarks of (Action) Role Shift in the strict sense, in particular body rotation, although they involve Role Shift in a broad sense (= the signer's body represents another character). The presence of first person possessive pronouns might suggest that no Role Shift is involved, but as shown in Appendix A, the deviance of such pronouns under Role Shift is less clear than expected. The presence of the agreement *RUN* with third person agreement is possibly a stronger argument that no Role Shift is involved, but the data are less than clear.

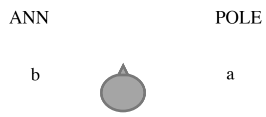
5 Adding Strict Role Shift

Our sentences with classifier movement conveying relative motion have counterparts with clear instances of Role Shift in the strict sense, involving in particular body rotation. Role Shift is traditionally thought to be associated with a character's viewpoint (Cormier et al. 2012, Quinto-Pozos and Parrill 2015, Waller 2021; Parrill, 2009, 2010). It is thus expected that Role Shift combined with moving characters can give rise to relative motion readings. For this reason, we discuss more briefly the case of dynamic viewpoints with Role Shift.

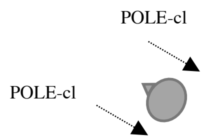
In (19), Action Role Shift is applied to the case, discussed in (7), in which Ann nearly hits a pole. Depending on whether the classifier passes the signer's head on his right or on his left, we understand that Ann nearly hit a pole or on her right or on her left.

(19) POSS-1 HOUSE HAVE POLE-a. SOMETIME CAR HIT-a. YESTERDAY ANN$_b$ SELF-b DRUNK.
'My house has a pole that cars sometimes hit. Yesterday, Ann was drunk, and
IX-b
she
a. clumsy__
7 RS$_b$______________________________________
DRIVE POLE POLE-passing_right-cl_ .
nearly hit the pole on her right while driving.'
b. clumsy__
RS$_b$______________________________________
7 DRIVE POLE POLE-passing_left-cl.
nearly hit the pole on her left while driving.'
(ASL 37, 3400)

(20) a. Loci for Ann and the pole nominal



b. Movement of the classifier representing the pole passing by in (19a) (top-right) and (19b) (bottom-left)



Similarly, (21), is a role-shifted version of (16). The initial loci for Ann and the tree are as in (22a) (which is identical to (18a)), but now Role Shift is applied to the last sentence, with a body rotation and a corresponding change in the path of the tree classifier moving towards the signer, as represented in (22b).

(21) YESTERDAY IX-1 SEE TREE TREE-cl$_a$, SELF PLANT TREE GROW. ANN PERSON-cl$_b$ SELF-b
'Yesterday I saw a tree I had planted myself. Ann

DRUNK. IX-b
was drunk. She
a. $_{\text{hurry}}$_
RS$_{a}$____
^{7}JOG TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
jogged towards the tree and hit her head against it.'
b. $_{\text{hurry}}$_
RS$_{a}$____[13]
$^{5.7}$ **b-RUN-a TREE TREE-cl-1$_{\textbf{moves and hits the signer's forehead}}$.**
ran towards the tree and hit her head against it.'
c. $_{\text{hurry}}$_
RS$_{a}$____
^{7} 1-RUN-front$_{\text{signer moves forward}}$ TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
ran towards the tree and hit her head against it.'
d. $_{\text{hurry}}$_
RS$_{a}$____
$^{6.7}$1-RUN-front$_{\text{signer doesn't move forward}}$ TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
ran towards the tree and hit her head against it.'
e. $_{\text{hurry}}$_
RS$_{a}$____
$^{5.3}$ **b-PERSON-cl-a TREE TREE-cl-1$_{\textbf{moves and hits the signer's forehead}}$.**
hurried towards the tree and hit her head against it.'
f. $_{\text{hurry}}$_
RS$_{a}$____
$^{6.7}$1-PERSON-cl-front$_{\text{signer moves forward}}$ TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
hurried towards the tree and hit her head against it.'
g. $_{\text{hurry}}$_
RS$_{a}$____
^{6} 1-PERSON-cl-front$_{\text{signer doesn't forward}}$TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
hurried towards the tree and hit her head against it.'
(ASL 37, 3154) https://youtu.be/HMGMdbDc_qg

(22) a. Loci for Ann and the tree classifier b. Movement of the tree classifier towards the signer

In all cases, movement of the tree classifier is of course interpreted as relative motion perceived by Ann. Example (21b) is a bit degraded, as expected because the subject agreement verb *RUN* pertains to Ann and is under Role Shift, but does not start from the signer's role-shifted position; the same remark applies to to (21e). What is a bit surprising is that deviance is not more marked. Specifically, it is not more marked than the counterpart without body shift in (16b), which only highlights the possibility that the latter might involve an incomplete version of Role Shift.

6 Movement of two classifiers relative to each other

All the examples discussed so far involved the motion of a single classifier relative to the signer, as in Lucas and Valli's constructions in (5). But Lucas and Valli also discuss constructions with two classifiers, as in (6), where a ground classifier moves relative to another classifier to represent relative motion. We sought to construct full paradigms with the same construction type. The result does not change the conceptual situation: a relative interpretation of movement is possible both with and without the body rotation

[13] Here our second consultant felt that the sentence is indeed suboptimal, due to misalignment with the Role Shift which should have less shoulder turn. See the Supplementary Materials.

that is often a hallmark of Role Shift. When there is no body rotation, the signer's head (and facial expressions) represent the character's head, and thus Role Shift in a broad sense is involved.

In (23), a person (index) classifier represents Ann running; the classifier involves repeated flexion of the finger articulation to evoke running, but the entire finger does not move in space. Simultaneously, a classifier representing a tree moves and represents the relative movement of trees from Ann's viewpoint. In (23a), the two classifiers are produced under Role Shift, with a clear body rotation, and effortful facial expressions. In (23b), no body rotation is present but the facial expressions persist.

(23) YESTERDAY ANN IX-b WOW ESCAPE POLICE. FOREST AREA-a IX-b
RS$_b$-effort____________________
a. ^{7}PERSON-cl$_{run}$^TREE-cl$_{passing}$-rep.
b. effort________________________
$^{6.7}$PERSON-cl$_{run}$^TREE-cl$_{passing}$-rep.
'Yesterday I saw Ann, she had to run away from the police. She passed several trees.'
(ASL 37, JL-24-11-25-2) https://youtu.be/LlKXzG730rE

A different example with the same structure appears in (24).

(24) YESTERDAY ANN(-b)[14] CAR RACE. IX-b MUST DRIVE PAST TELEPHONE POLE POLE-cl POLE-cl. IX-b
a. ^{7}RS$_b$-effort___________________
VEHICLE-cl^POLE-cl$_{passing}$-rep.
b. effort______________________
$^{6.7}$ VEHICLE-cl^POLE-cl$_{passing}$rep.[15]
'Yesterday Ann took part in a car race. She had to drive past telephone poles. She passed several poles.'
(ASL 37, JL-24-11-25-3) https://youtu.be/rup01ejppkM

7 Analysis I: variables denoting dynamic viewpoints

Since we arguably have examples both with and without Role Shift, we will develop the analysis in two steps. First, we will allow viewpoint variable to denote dynamic (i.e. moving) viewpoints. Second, we will adopt special measures for Role Shift. For reasons stated at the end of Section 4, all our examples involve Role Shift in the broad sense, which leaves open the possibility that only the second part of the theory is needed.

To analyze examples without Role Shift, we make a correction to the semantics of classifiers proposed in Schlenker and Lamberton 2024: instead of always being static, viewpoints will now have the ability to be dynamic. Technically, they will be treated as functions from times and worlds to static viewpoints (the temporal component is crucial, but we prefer to keep the parallel between times and worlds, hence the double dependency[16]).

To introduce the background, we start with dynamic classifiers interpreted relative to a *static* viewpoint. For Schlenker and Lamberton 2024, classifiers are expressions of the form P^{π}, where π is a viewpoint variable that tells the interpretive system that there is an iconic component, and simultaneously specifies the viewpoint relative to which this iconic component is interpreted. The lexical rule is given in its general form in (25a), and in the case of a static tree classifier in (25b) (we abstract

[14] In a., *ANN* was introduced in neutral location; in b., it was introduced in locus b.
[15] Here our second consultant feels that b. works but that its score should be lower; see the Supplementary Materials.
[16] To investigate world dependency, one would need to investigate modal cases involving the same viewpoint appearing in different positions in different worlds.

away from issues of trivalence). Here the symbolic elements (namely *P* and *TREE-cl*) must be tokens, not types, as their particular position in signing spaced is interpreted owing to the iconic component of the interpretive rule.

(25) a. $[[P^{\pi}]]^{c, s, t, w} = \lambda x_e \,.\, \boxed{[[P]]^{c, s, t, w}(x) = 1}$ and $\mathbf{proj(x, s(\pi), t, w) = \ P}$
b. $[[\text{TREE-cl}^{\pi}]]^{c, s, t, w} = \lambda x_e \,.\, \boxed{\text{tree}'_{t, w}(x) = 1}$ and $\mathbf{proj(x, s(\pi), t, w) = \ TREE\text{-}cl}$

This rule can be understood as follows. The boxed component is lexical, and the boldfaced component is iconic. The latter says that for an occurrence of P^{π} to hold true of an object x, x should project onto the symbol *P* from the viewpoint specified by π (hence $s(\pi)$ if the assignment function is s), at the time/world of evaluation (t and w).

We needn't go into the details of how projection works (this is discussed by way of a 2D pictorial semantics by Schlenker and Lamberton 2024, before they develop a 3D semantics appropriate for classifiers). But one assumption is essential for the present discussion: a viewpoint is in a fixed spatial position;[17] correspondingly, the *proj* notation should be understood as in (26), where the bold faced part ('from π') will be modified later to take into account the possibility that viewpoints move in space.

(26) **Projection with static viewpoints**
If d is an object, π a (static) viewpoint, t a time, w a world and P an occurrence of a classifier in signing space, proj(d, π, t, w) = P iff at t in w, d projects to P **from π**

With static viewpoints, classifiers can sometimes be dynamic. Schlenker and Lamberton give the following definition of dynamic projection, where the subscript τ indicates that the classifier is dynamic, i.e. moves in signing space. In a nutshell, the definition says that an object projects to a dynamic classifier just in case its various positions throughout a time interval given by the duration of the classifier (modulo a scaling component) project to the corresponding positions of the classifier.

(27) **Dynamic projection** (simplified[18])
Let ρ^* be the Cartesian frame of reference associated with the signer, and let $\textit{word-cl}_{\tau}$ be a dynamic predicate classifier. If δ is the duration of the classifier movement, we view $\textit{word-cl}_{\tau}$ as a function from the interval $[0, \delta]$ to static classifiers. If π is a viewpoint, $\tau(\pi)$ a temporal scaling factor determined by π, t is a time, w a world and d and entity:
$\text{proj}(\pi, t, w, d) = \text{word-cl}_{\tau}$ iff for each $\delta' \leq \delta$, $\text{proj}(d, \boldsymbol{\pi}, t+(\tau(\pi))\delta', w) = \text{word-cl}_{\tau}(\delta')$.

To illustrate, consider the example in (28), where a person classifier representing Obama moves towards the signer for 3 seconds.

(28) $\text{proj}(\pi, t, w, \text{Obama}) = \text{person-cl}_{\tau}$

Applying the notations of (27) to this case, $\text{word-cl}_{\tau} = \text{person-cl}_{\tau}$, d = Obama, and δ, the duration of the classifier movement, is 3s. Suppose further that the viewpoint π (corresponding to the camera position) is a journalist, and that the scaling factor $\tau(\pi)$ is 10. The condition in (28) will be satisfied just in case Obama moves towards the journalist for 30 seconds at the time t and in the world w of evaluation.[19]

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[17] More precisely, Schlenker and Lamberton 2024 specify that "any viewpoint π specifies **a Cartesian frame of reference $\boldsymbol{\rho(\pi)}$**, a spatial scaling factor $\sigma(\pi)$ (and for dynamic projection, a temporal scaling factor $\tau(\pi)$).". The boldfaced part implies that the frame of reference is given and thus does not move.
[18] This version omits the condition that the object d should fall under the lexical predicate *word*; but as discussed in Schlenker and Lamberton 2024, this is redundant with the condition in (25).
[19] In greater detail: the condition (i) from (27) can be rewritten as in (ii) and paraphrased as in (iii).
(i) for each $\delta' \leq \delta$, $\text{proj}(d, \pi, t+(\tau(\pi))\delta', w) = \text{word-cl}_{\tau}(\delta')$
(ii) for each $\delta' \leq 3s$, $\text{proj}(\text{Obama, the journalist}, t+10\delta', w) = \text{person-cl}_{\tau}(\delta')$

To take into account the existence of moving viewpoints, we can keep the very same notations, but interpret them differently. A viewpoint will henceforth be something that, just like a person, can be in different places at different times and in different possible worlds, hence the modification of the boldfaced part in (29).

(29) **Projection with dynamic viewpoints**
If d is an object, π a (dynamic) viewpoint, t a time, w a world and P an occurrence of a classifier in signing space, proj(d, π, t, w) = P iff at t in w, d projects to P **from π(t, w), the static viewpoint specified by π at t in w**

In other words, a dynamic viewpoint is now a function from times and worlds to static viewpoints; it is immediate that the case of standard (non-moving) viewpoints is a special case of this more general framework: the function is constant.

To illustrate, consider our earlier example in (30), whose last word is analyzed as in (31), complemented by (32).

(30) ANN PERSON-cl$_a$ SELF-b DRUNK. IX-b JOG TREE TREE-moves and hits the signer's forehead-cl-1
'Ann was drunk. She jogged towards the tree and hit her head against it.'

(31) a. TREE-cl$^\pi$(b)
b. $[[\text{TREE-cl}_\tau{}^\pi(b)]]^{c,s,t,w} = [\lambda x_e . \text{tree}'_{t,w}(x) = 1$ and proj(x, s(π), t, w) = tree-cl$_\tau$](s(b)
= [tree'$_{t,w}$(s(b)) = 1 and proj(s(b), s(π), t, w) = tree-cl$_\tau$]

(32) If δ is the duration of the movement of the classifier TREE-cl$_\tau$:
proj(s(b), s(π), t, w) = tree-cl$_\tau$ iff for each $\delta' \leq \delta$, proj(s(b), π, t+($\tau(\pi)$)δ', w) = tree-cl$_\tau$(δ')
iff for each $\delta' \leq \delta$, at t+($\tau(\pi)$)δ' in w, s(b) projects to tree-cl$_\tau$(δ') from viewpoint **π(t+($\tau(\pi)$)δ', w)**.

On the assumption that the tree position remains fixed, if the tree classifier moves towards the signer for 1 second and the scaling factor $\tau(\pi)$ is 100, this requires that the viewpoint (i.e. Ann) moves towards the tree for 100 seconds or 1m40s.[20]

The same analysis can be applied to an example in which a ground classifier moves relative to another classifier. The dynamic viewpoint is now anchored to the character (represented by the signer's face). For instance, for a ground classifier moving relative to a vehicle classifier, the dynamic viewpoint may follow the vehicle, so that from that perspective the vehicle doesn't move (note that the dynamic viewpoint is unlikely to be inside the vehicle, or else the vehicle itself couldn't be displayed, or at least not in the same way).

(iii) for each $\delta' \leq 3$s, from the journalist's fixed viewpoint, in w, at time t+10δ', Obama projects to whatever position the person classifier has in signing space at time δ'.
In particular, if the Obama classifier starts at the right of signing space and reaches the center of signing space 3 seconds later, at t Obama must project to the right of signing space, and at t+30s Obama must project to the center of signing space.

[20] In greater detail: the condition (i) from (32) can be rewritten as in (ii) and paraphrased as in (iii), taking s(b) to be the tree, $\tau(\pi)$ to be 100, and π(t+($\tau(\pi)$)δ') to be Ann's viewpoint at t+($\tau(\pi)$)δ'.
(i) for each $\delta' \leq \delta$, proj(s(b), π(t+($\tau(\pi)$)δ'), t+($\tau(\pi)$)δ', w) = tree-cl$_\tau$(δ')
(ii) for each $\delta' \leq 1$s, proj(the tree, π(t+100δ'), t+100δ', w) = tree-cl$_\tau$(δ')
(iii) for each $\delta' \leq 1$s, from Ann's moving viewpoint at t+100δ', in w, at time t+100δ', the tree projects to whatever position the person classifier has in signing space at time t+100δ'.
In particular, if the tree classifier starts out in the front of signing space and hits the signer's forehead 1 second later, then at t the tree must project to the front of signing space from Ann's viewpoint at t, which means that Ann must be somewhat far from the tree; while at t+100s the tree must project to the back of signing space from Ann's viewpoint at t+100s, which means that the tree must be in contact with Ann.

8 Analysis II: dynamic viewpoints introduced by Role Shift

We turn to the analysis of relative motion under Action Role Shift. Intuitively, Role Shift presents actions from the character's perspective, but this idea has not been fully cashed out in terms of a pictorial semantics. We propose an analysis within an earlier theory of Role Shift as overt context shift (Schlenker 2017a, b).

8.1 Dynamic viewpoints and the context-shifting analysis of Role Shift

Within formal semantics, one view of Role Shift, suggested by Quer 2005, 2013 and implemented in Schlenker 2017a,b, is that it is an overt instance of context shift. This view was proposed by Quer 2005 for Attitude Role Shift in Catalan Sign Language (LSC) because it involves instances of indirect discourse in which indexical expressions such as *I* can refer to the agent rather than the signer.[21] Schlenker 2017a extended a version of this analysis to ASL by positing that Role Shift shifts the context of evaluation of indexicals. In Attitude Role Shift, the shifted context is a context of speech or thought, and indexicals refer to the character, not to the signer. In Action Role Shift, the shifted context is just a perspectival point.

Without getting into full details, we can sketch an extension of this analysis to incorporate viewpoints. We propose to add a distinguished viewpoint variable, π^*, which depends on the context: it denotes the viewpoint associated with the agent of the context. The distinction between normal (unstarred) viewpoint variables and context-bound viewpoint variables is defined in (33).

(33) a. $[[\pi_i]]^{c, s, t, w} = s(\pi_i)$
b. $[[\pi^*]]^{c, s, t, w} = \pi(c_a)$, the **dynamic** viewpoint associated with the agent of c

Importantly, $\pi(c_a)$ in (33b) is a dynamic viewpoint (technically: a function from time-world pairs to static viewpoints), and correspondingly we must make use of the revised definition of projection given in (29): if projection is evaluated at t in w, the (static) viewpoint of evaluation will be $\pi(c_a)(t, w)$, corresponding to the viewpoint of individual c_a at t in w. Regarding $s(\pi_i)$ in (33), there is a choice. On the more restrictive version of the theory, $s(\pi_i)$ is a constant function from time-world pairs to static viewpoints—i.e. a static viewpoint. On this view, only Role Shift can introduce dynamic viewpoints. On a more liberal theory, $s(\pi_i)$ may be a non-constant function from time-world pairs to static viewpoints, which allows for the extensions discussed in Section 7.

In the framework of Schlenker 2017a, the Role Shift operator triggers abstraction over contexts, as is defined in (34) and illustrated in (35) (context change will matter for the interpretation of the viewpoint variable π^* in (33b), as it depends on the context parameter). For reasons of simplicity, Schlenker 2017a only takes into account world dependency, but for present purposes it is important that we add time dependency; we treat times by analogy with worlds.

(34) Let c be a context, s an assignment function, t a time and w a world. Then for any index i and clause *IP*,
$[[RS_i\ IP]]^{c, s, t, w} = \lambda x'_e\ \lambda t'_i\ \lambda w'_s\ .\ [[IP]]^{<x', t', w'>, s, t', w'}$

(35) Let c be a context, s an assignment function, t a time and w a world.
$[[RS_a\ [\text{IX-1 WILL-LEAVE}]]]^{c, s, t, w}$
$= \lambda x'_e\ \lambda t'_i\ \lambda w'_s.\ [[\text{IX-1 WILL-LEAVE}]]^{<x', t', w'>, s, t', w'}$

[21] The argument that indirect rather than direct discourse was involved rested on the presence in the target clause of other indexicals, such as *here*, which were evaluated with respect to the signer. Thus in was is literally 'When he was in Madrid, Joan thought that I would finish my study here', *here* refers to the signer's location (say, Barcelona), while *I* refers to Joan.

$= \lambda x'_e\, \lambda t'_i\, \lambda w'_s.\ [\![\text{WILL-LEAVE}]\!]^{<x', t', w'>, s, t', w'}([\![\text{IX-1}]\!]^{<x', t', w'>, s, t', w'})$
$= \lambda x'_e\, \lambda t'_i\, \lambda w'_s.\ \text{will-leave}'_{t', w'}(x')$

When the role-shifted element contains a classifier with a context-bound variable π^*, Role Shift makes its effects felt, as illustrated in the expression in (36a), analyzed with the Logical Form in (36a') and the truth conditions in (36b). Here we assume that that the tree classifier, which is a predicate, takes as argument a locus *b*, which is a variable denoting the relevant tree.

(36) a. RS_a b-TREE-cl_τ
a'. RS_a TREE-$cl_\tau^{\pi^*}$(b)
b. Let c be a context, s an assignment function, t a time of evaluation and w a world of evaluation.
$[\![\text{(a')}]\!]^{c, s, t, w} = \lambda x'_e\, \lambda t'_i\, \lambda w'_s.\ [\![\text{TREE-cl}_\tau^{\pi^*}(b)]\!]^{<x', t', w'>, s, t', w'}$
$= \lambda x'_e\, \lambda t'_i\, \lambda w'_s.\ [\text{tree}'_{t', w'}(s(b)) = 1 \text{ and proj}(s(b), \boxed{\pi(x')}, t', w') = \text{tree-cl}_\tau]$

In the boxed part of (36), π(x') is the viewpoint associated with the individual x'. Importantly, this viewpoint is dynamic and thus takes different values at different times.

We can then apply this analysis to a full clause that includes a moving tree classifier, as in (37a). We follow Schlenker 2017a in positing a covert world variable *w** in the Logical Form, as in (37b), to which we add a covert time variable *t** (since we treat times by analogy with worlds).

(37) RS_a____________
a. IX-a b-TREE-move-cl

b. w* t* IX-a RS_a [TREE-move-cl^{π^*}(b)]

c. $[\![(b)]\!]^{c, s, t, w} = [\lambda x'_e\, \lambda t'_i\, \lambda w'_s.\ [\![\text{TREE-cl}_\tau^{\pi^*}(b)]\!]^{<x', t', w'>, s, t', w'}](s(a))(s(t^*))(s(w^*))$
$= [\![\text{TREE-cl}_\tau^{\pi^*}(b)]\!]^{<s(a), s(t^*), s(w^*)>, s, s(t^*), s(w^*)}$
$= \text{tree}'_{t', w'}(s(b)) = 1 \text{ and } \boxed{\text{proj}(s(b), \pi(s(a)), s(t^*), s(w^*)) = \text{tree-move-cl}_\tau}]$

In words: the denotation s(b) of locus b is a tree and dynamically projects to the classifier *TREE-move-cl* from the viewpoint specified by the denotation s(a) of locus a at the time and world s(t*) and s(w*) specified by the variables *t** and *w**.

d. Let us write π(s(a)) as π_a, s(t*) as T, s(w*) as W. Using (32), we can expand the boxed part of c. as follows:
$\text{proj}(s(b), \pi(s(a)), T, W) = \text{tree-cl}_\tau$ iff for each $\delta' \leq \delta$, $\text{proj}(s(b), \pi_a, T+(\tau(\pi_a))\delta', W) = \text{tree-cl}_\tau(\delta')$,
iff for each $\delta' \leq \delta$, at $T+(\tau(\pi_a))\delta'$ in w, s(b) projects to $\text{tree-cl}_\tau(\delta')$ from viewpoint $\boldsymbol{\pi_a(T+(\tau(\pi_a))\delta', W)}$
In words: s(b) dynamically projects to the classifier *TREE-move-cl* from dynamic viewpoint π(s(a)) at T in W just in case at every moment of a time interval given by the duration of the classifier modulo scaling, s(b) projects to the position of the classifier from the value of the viewpoint at the relevant moment.

As in our earlier examples, on the assumption that the tree (i.e. s(b)) does not move, the classifier movement provides information about the movement of the viewpoint.

8.2 Dynamic viewpoints and the demonstrative analysis of Role Shift

An alternative analysis of Role Shift is developed in Davidson 2015, who follows Supalla 1982 in taking Role Shift to involve a "body classifier", in the sense that the signer herself demonstrates aspects of another character's speech/thought or actions. In Action Role Shift, the role-shifted verb refers (self-referentially) to its own form, which accounts for its iconic implications. For instance, a verb (or classifier) meaning *leave* signed under Role Shift while co-occurring with a happy face means in essence *left like this*, and the happy face must be attributed to the character. As it stands, this theory has no articulated pictorial semantics and in particular no notion of viewpoint, which makes it difficult to sharpen the issue of dynamic viewpoints. It also encounters a more substantive difficulty: the analysis was explicitly designed to handle only cases in which a single verb (rather than entire clause) appears under Action Role Shift. But in all our examples, a full clause appears under Action Role Shift, as can be seen in (21), for instance. We leave it for future research to determine how dynamic viewpoints could be integrated to this analysis.

9 Conclusion

As emphasized by Liddell 2003 and especially Müller 2018 (see also Parrill et al. 2016), sign language can make use of cinematic techniques, here of traveling cameras—which translate into moving viewpoints in our examples. In the simplest analysis, this requires a tweak to Iconological Semantics as developed in Schlenker and Lamberton 2024, but it is a natural tweak: viewpoints should be dynamic (technically, they should be functions from time-world pairs to static viewpoints). But there are also cases in which dynamic viewpoints are introduced by Role Shift. Within the context-shifting analysis of Role Shift, we extended our theory to allow a distinguished viewpoint variable to depend on the context parameter.

With a strict definition (involving in particular body shift), only our examples of Section 5 displayed Role Shift. With a broad definition (involving any case in which the signer's face/body represents another character), all our examples displayed Role Shift. Thus if one adopts the latter definition, one could pursue a restrictive theory on which only Role Shift can introduce dynamic viewpoints.

This is not just a terminological question. The key issue is whether the cluster of properties associated with Role Shift in ASL, involving in particular indexical pronouns and agreement verbs, correlates with the appearance of dynamic viewpoints. We started to test these correlations, but the data are less than clear and will require further research. We thus leave open whether dynamic viewpoints can be introduced by all viewpoint variables, or only by those associated with Role Shift (see also Waller 2021 for complexities in the definition of Role Shift). We also leave for future research a refinement of the analysis of Viewpoint Shift in relation to Role Shift. In addition, it is clear that our data should be assessed with further consultants in the future.

Be that as it may, it is a remarkable fact that the device of traveling cameras has a direct counterpart in natural language. And it is reassuring that this fact can be naturally integrated to a formal framework independently developed to analyze the interaction between logic and iconicity in sign language.[22]

[22] One last but important point pertains to Iconological Semantics itself. Some of Schlenker and Lamberton's (2024) arguments for viewpoint variables depend in part on the assumption that viewpoints are *not* functional. For instance, there is no need for separate viewpoint variables in (1a) if viewpoints are time-dependent: a single viewpoint located in different classrooms at different times may suffice. Relatedly, see Schlenker and Lamberton 2024, Appendix I-C and I-D, for a discussion of the possibility of eliminating existential quantification over viewpoints in favor of functional viewpoints.

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Philippe Schlenker: Institut Jean-Nicod (ENS - EHESS - CNRS), Département d'Etudes Cognitives, Ecole Normale Supérieure, Paris, France; PSL Research University. philippe.schlenker@gmail.com
Jason Lamberton: Motion Light Lab, VL2, Gallaudet University, USLA. jason.lamberton@gmail.com
Jonathan Lamberton: New York, NY, USA. jonlamberton@gmail.com

Appendix

A. Elicitation Methods and Transcription Conventions

The main consultant (and co-author) is a Deaf, native signer of ASL (of Deaf, signing parents).[23] Elicitation was conducted using the 'playback method', described for instance in Schlenker et al. 2013, Schlenker and Lamberton 2019, 2022. It involved repeated quantitative acceptability judgments (on a 7-point scale with 7 = best), as well as inferential judgments. Average acceptability for our main consultant appears as a superscript at the beginning of sentences. Raw scores appear when there was more than a 2-point difference on the same question across different sessions. References such as (e.g. *ASL 37, 3126, 3 judgments*) at the end of paradigms cross-reference the ASL video (here video 37, 3126) and indicate the number of iterated judgment tasks (on different days, here three). We provide links to anonymized versions of ASL videos, and specialists are invited to consult the raw judgments in the Supplementary Materials when relevant. After the data were collected, Jonathan Lamberton informally checked the judgments reported for original data with his brother Jason Lamberton (also a co-author), who is a Deaf native signer of ASL as well; the main points of disagreements are noted as we go (in footnotes when they are inessential), and a full description appears in the Supplementary Materials. Unless noted otherwise, the two consultants agreed with respect to the contrasts we report.

Transcription conventions are standard for sign language. Loci are alphabetized from dominant to non-dominant side from the signer's perspective (here: from right to left). A suffixed locus, as in *WORD–i,* indicates that the word points toward locus i (a position of signing space associated with a discourse referent). *IX-i* (for 'index') is an index sign pointing toward locus *i*. $EXPRESSION_i$ is used for expressions associated with locus *i* by virtue of being signed in (rather than by pointing to) the corresponding area of signing space. Agreement verbs include loci in their realization—for instance, the verb *a-HIT-b* starts out from locus *a* and targets locus *b*. We put *-cl* at the end of classifier predicates, and describe their movement with affixed words (e.g. *TREE-moving on the signer's left-cl*). For clarity, we sometimes append a picture of the classifier right after its transcription. Among non-manuals, only Role Shift is transcribed, as *RS*, followed by a line indicating its extent (additional non-manuals are notated with *ad hoc* descriptions, using the same line-based notation when relevant, sometimes with *RS* if coextensive with it).

B. Possessive First Person Pronouns under Action Role Shift

Earlier work argued that under Role Shift, ASL first person pronouns cannot refer to the signer (Schlenker 2017a). This is for two separate reasons depending on the type of Role Shift involved. In Attitude Role Shift, which serves to report somebody's thoughts or words, first person pronouns are acceptable but denote the attitude holder, not the signer. In Action Role Shift, which serves to represent actions in a particularly vivid fashion, first person pronouns are unacceptable (though first person agreement markers are acceptable). None of our examples involve attitude reports, hence only Action Role Shift is relevant here, and we expect first person pronouns to be degraded. However judgments are unstable and mixed.

The example in (38) is similar to the role-shifted example (19) from the main

[23] We use the term *consultant* to refer to a collaborator that assesses sentences, including if this person is also a contributor to the article.

text, except that the first person possessive *POSS-1* is added under Action Role Shift (for this reason, it is also role-shifted version of (7) from the main text). Acceptability is less high than for (19), but the contrast is unclear. The signer's facial expressions change after the verb *DRIVE*, which might suggest that Role Shift is interrupted, and our main consultant sees a "slight release" of Role Shift on the possessive pronoun. But this is non-trivial to reconcile with the fact that body rotation is not interrupted.

(38) POSS-1 HOUSE HAVE POLE-a. SOMETIME CAR HIT-a. YESTERDAYANN$_b$ SELF-b DRUNK.
'My house has a pole that cars sometimes hit. Yesterday, Ann was drunk, and
IX-b
she
a. clumsy_
RS$_b$____
$^{5.7}$ DRIVE POSS-1 POLE POLE-passing_right-cl_.
nearly hit the pole on her right while driving.'
b. clumsy__
RS$_b$____
$^{5.7}$ DRIVE POSS-1 POLE POLE-passing_left-cl.
nearly hit the pole on her left while driving.'
(ASL 37, 3402) https://youtu.be/LHYPzH8pt9g

By the same token, (39) is similar to the role-shifted example (21) from the main text, except that *POSS-1* is once again added under Action Role Shift (this is also a role-shifted version of (16) from the main text). Acceptability judgments are in part unstable, and some of the examples are somewhat degraded. This might suggest that first person possessive pronouns are degraded under Role Shift. To the extent that they are not, it can be noted that a backward nod accompanies the possessive pronoun, which might suggest that Role Shift is somehow interrupted; but this is non-trivial to reconcile with the fact that body rotation continues. Here our second consultant's acceptability judgments were rather different from those of our main consultant, and his numerical values appear within square brackets (e.g. in (39)a, the notation $^{5.3\,[4]}$ indicates that the average score for our main consultant is 5.3, and the score that our second consultant mentioned in the informal description of his judgment is 4). Our second consultant's scores suggest that all examples are a bit or very degraded.

(39) YESTERDAY IX-1 SEE TREE TREE-cl$_a$, SELF PLANT TREE GROW. ANN PERSON-cl$_b$ SELF-b
'Yesterday I saw a tree I had planted myself. Ann
DRUNK. IX-b
was drunk. She
a. backward nod_______[24]
RS$_a$____
$^{5.3\,[4]}$ JOG POSS-1 TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
jogged towards the tree and hit her head against it.'
b. backward nod_______
RS$_a$____
$^{5\,[4]}$ b-RUN-a POSS-1 TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$. (3, 6, 6)
ran towards the tree and hit her head against it.'
c. backward nod_______
RS$_a$____
$^{5.7\,[3]}$ 1-RUN-front$_{\text{signer moves forward}}$ POSS-1 TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
ran towards the tree and hit her head against it.'

[24] The backward nod is a return to neutral posture (while retaining body shift). Correspondingly, on some sentences the backward nod seems stronger, as in c. and f., where the signer initially moves forward; it is more subtle in b. and e., where the signer doesn't vary move forward much when displaying the running.

d. backward nod

RS$_a$

5.7 [3] 1-RUN-front$_{\text{signer doesn't move forward}}$ POSS-1 TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$. (4, 7, 6)
ran towards the tree and hit her head against it.'

e. backward nod

RS$_a$

5 [5] b-PERSON-cl-a POSS-1TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$. (3, 6, 6)
hurried towards the tree and hit her head against it.'

f. backward nod

RS$_a$

5.7 [4] 1-PERSON-cl-front$_{\text{signer moves forward}}$ POSS-1 TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
hurried towards the tree and hit her head against it.'

g. backward nod

RS$_a$

5 [4] 1-PERSON-cl-front$_{\text{signer doesn't move forward}}$ POSS-1 TREE TREE-cl-1$_{\text{moves and hits the signer's forehead}}$.
hurried towards the tree and hit her head against it.'
(ASL 37, 3152) https://youtu.be/NxurDjnnebg

Similarly, we added Role Shift to the non-role-shifted example (10) from the main text, which includes a first person possessive. The result, in (40), gave rise to unstable judgments, and not all examples are significantly degraded. Here too, there is a change of facial expressions after *DRIVE*, which might suggest that Role Shift is interrupted, except for the crucial fact that body rotation is not interrupted. Our second consultant's judgments differed once again from our main consultant's, and we thus include the former's scores within square brackets. Our second consultant's scores suggest that all examples are a bit or very degraded.

(40) YESTERDAY IX-1 SEE POSS-1 HOUSE-a ANN SELF-b DRUNK. IX-b DRIVE
'Yesterday I saw my house, and nearby Ann, who was drunk. She

RS$_b$

b-DRIVE-cl-mid POSS-1 HOUSE$_a$ mid-DRIVE-cl-a-a$_{\text{front}}$
drove towards my house and took a left, nearly missing

a.
6 [3] WALL-cl-1$_{\text{signer's right}}$.
my house on her right.'

b.
5.5 [4] RECTANGLE-cl-1$_{\text{signer's right}}$.
my house on her right.'

c.
6 [3] CORNER-cl-1$_{\text{signer's right}}$.
the corner of my house on her right.'

d.
4.5 [5] WALL-cl-1$_{\text{signer's left}}$. (5, 6, 4, 3)
my house on her left.'

e.
4.8 [5] RECTANGLE-cl-1$_{\text{signer's left}}$.
my house on her left.'

f.
4.5 [4] CORNER-cl-1$_{\text{signer's left}}$. (5, 6, 4, 3)
the corner of my house on her left.'
(ASL 37, 3382, 4 judgments) https://youtu.be/cZlT1gMLB6I

In sum, on the basis of these and further examples we tested, we were unable to clearly display the expected deviance of first person possessive pronouns under Action Role Shift. Still, the standard prohibition against first person pronouns under Action Role Shift might not be wrong, as suggested by two facts. First, most examples were degraded for our second consultant. Second, examples that were somewhat acceptable for our main consultant often included a subtle interruption of Role Shift.

C. ASL Raw Data: Supplementary Materials and Anonymized Videos

Supplementary materials regarding ASL raw data can be found in the following file: https://osf.io/ya9nk/

It contains:
1. all of our main consultant's detailed judgments, as well as a summary of our second consultant's judgments;
2. averages computed from an Excel file;
3. anonymized videos.